

Shadman Salif Swanan

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EDUCATION

BRAC University

Jan. 2022 – Present

Bachelor of Science in Computer Science and Engineering (CSE)

Dhaka, Bangladesh

- **CGPA:** 3.82 out of 4.00
- **Programming Coursework:** Fundamentals of Programming, Object-Oriented Programming, Data Structures & Algorithms
- **Computational Coursework:** Natural Language Processing, Linear Algebra, Neural Networks, Digital Logic Design, VLSI Design, Discrete Mathematics
- **Programming Languages:** Python, Java, C/C++, JavaScript, TypeScript
- **Frameworks & Tools:** Git, OpenGL, MySQL, React, Next.js, MERN Stack, Figma, Detectron2

PROJECTS

BracBalance

- Banking and finance management application designed to track expenses and manage personal financial records
- Implemented core transaction logic and data handling features to ensure accurate balance tracking

Natural Language Processing POS Tagging

- Implemented token-level Part-of-Speech tagging using multiple RNN-based architectures on the POS1 dataset
- Compared performance of different recurrent models for sequence labeling tasks

Shikkha.com

- Developed an Ed-Tech platform using JavaScript to support digital learning and content delivery
- Designed modular front-end components for scalability and user interaction

Cafeteria Management System

- Built a management system to automate cafeteria operations including order handling and record management
- Improved operational efficiency through structured data storage and processing

Voting System (x86 Assembly)

- Designed and implemented a secure voting system interface using emu8086 and x86 Assembly
- Included voter authentication, eligibility verification, and vote recording mechanisms

3D Car Racing Game

- Developed a 3D car racing game using OpenGL featuring player controls, enemy vehicles, and environmental effects
- Implemented game mechanics including movement physics, collision logic, and camera control

RESEARCH EXPERIENCE

Undergraduate Thesis Research

Sept. 2025 – Present

BRAC University

Dhaka, Bangladesh

- Conducting research on **semantic drift in the Bengali language** using diachronic word embeddings and transformer-based representations
- Reviewed and analyzed prior literature on semantic change detection, historical embeddings, and contextual language models
- Developing a column detection model for scanned Bengali newspapers to improve document layout understanding
- Building an OCR pipeline to construct a clean and high-quality diachronic text corpus for downstream NLP tasks

CO-CURRICULAR ACTIVITIES

Notre Dame Nature Study Club

Vice President, Department of Photography

- Led photography initiatives and coordinated visual documentation for club activities and events
- Managed team members and contributed to creative direction and event coverage

Ideal English Language Club

General Member

- Assisted in organizing language-focused programs, workshops, and internal club operations
- Contributed to communication management and event coordination

BRAC University Residential Semester

Stage Drama Course Participant

- Completed an intensive stage drama course emphasizing teamwork, communication, and performance skills

BRAC University Sports Activities

Football Achievements

- Runners-Up, RS Cup
- Runners-Up, Residential Semester Football Tournament
- Champion and Top Scorer, Intra-Spring Football Tournament, BRAC University